

IES/ISS Statistics Paper-I

Probability: Data-Mined 2027 Forecast

Topic-wise prediction from the supplied 2018–2026 mine

Forecast Basis

This forecast uses the supplied classification of 196 Probability questions from nine papers. The numerical values below are empirical forecast probabilities for a question archetype or syllabus block to appear in the 2027 Probability portion. They are not official probabilities and do not constitute an answer key for any previous paper.

The principal signals used are:

forecast strength = historical frequency+recent momentum+multi-question bundling+uncovered syllabus pressure.

The strongest structural signal is the repeated use of one joint distribution to generate two or three questions. The highest-value preparation therefore remains:

$$P6 > P8 > P5 \approx P9 > P4 > P7 > P2 > P1 > P3 > P12 \approx P13 > P10 > P11.$$

Predicted 2027 Allocation

Code	Sub-topic	Forecast questions	Topic hit	Priority
P1	Classical and axiomatic probability	1	0.78	High
P2	Conditional probability and Bayes' theorem	1	0.82	High
P3	Random variables and distribution functions	1	0.67	Medium
P4	Standard discrete distributions	3	0.98	Very high
P5	Standard continuous distributions	3	0.96	Very high
P6	Random vectors and joint distributions	5	0.995	Exceptional
P7	Distributions of functions of random variables	2	0.93	Very high
P8	Mathematical and conditional expectation	3	0.97	Very high
P9	Characteristic functions, MGF and PGF	3	0.94	Very high
P10	Modes of convergence	1	0.56	Medium
P11	Borel and Kolmogorov 0-1 laws	1	0.34	Low but examinable
P12	Tchebycheff and Kolmogorov inequalities	1	0.58	Medium
P13	Laws of large numbers and CLT	1	0.70	Medium
Total forecast		26		

Interpretation. The expected 2027 Probability load is approximately 22–27 questions, with the central forecast at 26. P6 alone is expected to contribute approximately 4–6 questions, frequently through one shared joint-density stem.

P1: Classical and Axiomatic Probability

Hit probability: 0.58. *Signal:* Geometric probability, discriminant conditions, and independence-based complements.

Forecast Problem 1. Let U and V be selected independently and uniformly from $(0, 1)$. What is the probability that the equation

$$t^2 - Ut + V = 0$$

has real roots?

- (a) $\frac{1}{24}$
- (b) $\frac{1}{10}$
- (c) $\frac{1}{12}$
- (d) $\frac{1}{8}$

P2: Conditional Probability and Bayes' Theorem

Hit probability: 0.76. *Signal:* Unequal-prior factory or agency Bayes template.

Forecast Problem 2. Factories A , B , and C produce 50%, 30%, and 20% of the total output, respectively. Their defective rates are 1%, 2%, and 4%, respectively. Given that a randomly selected item is defective, the probability that it was produced by factory C is

- (a) $\frac{4}{19}$
- (b) $\frac{8}{19}$
- (c) $\frac{10}{19}$
- (d) $\frac{12}{19}$

P3: Random Variables and Distribution Functions

Hit probability: 0.51. *Signal:* Piecewise CDF, endpoint continuity, and admissible parameter detection.

Forecast Problem 3. The function

$$F(x) = \begin{cases} 0, & x < 0, \\ \frac{x}{3}, & 0 \leq x < 1, \\ \alpha x + \frac{1}{3} - \alpha, & 1 \leq x < 2, \\ 1, & x \geq 2 \end{cases}$$

is the distribution function of a continuous random variable if and only if

- (a) $0 \leq \alpha \leq \frac{1}{3}$
- (b) $\frac{1}{3} \leq \alpha < \frac{2}{3}$

(c) $\alpha = \frac{2}{3}$

(d) $0 < \alpha < 1$

P4: Standard Discrete Distributions

Hit probability: 0.89. *Signal:* Binomial mode and skewness, negative-binomial waiting times, and hypergeometric variance.

Forecast Problem 4. Let $X \sim \text{Binomial}(12, 1/4)$. Consider the following statements:

I. The mode of X is 3.

II. The distribution of X is positively skewed.

Which of the statements is or are correct?

(a) I only

(b) II only

(c) Both I and II

(d) Neither I nor II

Forecast Problem 5. A family continues to have children until its second daughter is born. Assuming that the probabilities of a male and a female child are both $1/2$, what is the probability that the second daughter is the seventh child?

(a) $\frac{3}{64}$

(b) $\frac{5}{64}$

(c) $\frac{15}{128}$

(d) $\frac{21}{128}$

Forecast Problem 6. A population contains 8 marked and 12 unmarked objects. A sample of 5 objects is drawn without replacement. If X denotes the number of marked objects in the sample, then $\text{Var}(X)$ is

(a) $\frac{9}{19}$

(b) $\frac{12}{19}$

(c) $\frac{15}{19}$

(d) $\frac{18}{19}$

P5: Standard Continuous Distributions

Hit probability: 0.86. *Signal:* Beta parameter recovery, exponential interval optimization, and Cauchy moment traps.

Forecast Problem 7. A beta distribution of the first kind has mean $1/3$ and variance $1/45$. Its parameters are

- (a) 2, 4
- (b) 3, 6
- (c) 4, 8
- (d) 6, 3

Forecast Problem 8. Let X have the exponential distribution with rate parameter $\theta > 0$. The maximum possible value of

$$\mathbb{P}(3 \leq X \leq 6)$$

over all $\theta > 0$ is

- (a) $\frac{1}{8}$
- (b) $\frac{1}{6}$
- (c) $\frac{1}{4}$
- (d) $\frac{1}{2}$

Forecast Problem 9. If X has the standard Cauchy distribution, which of the following is correct?

- (a) Both $\mathbb{E}(X)$ and $\text{Var}(X)$ exist.
- (b) $\mathbb{E}(X)$ exists but $\text{Var}(X)$ does not exist.
- (c) $\text{Var}(X)$ exists but $\mathbb{E}(X)$ does not exist.
- (d) Neither $\mathbb{E}(X)$ nor $\text{Var}(X)$ exists.

P6: Random Vectors, Joint, Marginal and Conditional Distributions

Hit probability: 0.95 for at least one question; **Hit probability:** 0.78 for a shared two-question or three-question

Signal: This is the dominant 2027 block. Sketching the support before integrating is essential.

Forecast Problem 10. Let the joint density of X and Y be

$$f(x, y) = \begin{cases} x + y, & 0 < x < 1, 0 < y < 1, \\ 0, & \text{otherwise.} \end{cases}$$

Then

$$\mathbb{E}(Y \mid X = 1/2)$$

is equal to

- (a) $\frac{5}{12}$
- (b) $\frac{7}{12}$
- (c) $\frac{2}{3}$
- (d) $\frac{3}{4}$

Forecast Problem 11. Let

$$f(x, y) = \begin{cases} 2, & 0 < x < y < 1, \\ 0, & \text{otherwise.} \end{cases}$$

The marginal density of X is

- (a) $f_X(x) = 2x, \quad 0 < x < 1$
- (b) $f_X(x) = 2(1 - x), \quad 0 < x < 1$
- (c) $f_X(x) = 1 - x, \quad 0 < x < 1$
- (d) $f_X(x) = 1 + x, \quad 0 < x < 1$

Forecast Problem 12. Let

$$f(x, y) = \begin{cases} \frac{21}{2}x^2y, & 0 < x < 1, \ x^2 \leq y, \\ 0, & \text{otherwise.} \end{cases}$$

Then

$$\mathbb{P}(X \geq Y)$$

is equal to

- (a) 0.12
- (b) 0.25
- (c) 0.30
- (d) 0.70

Forecast Problem 13. Suppose

$$(X_1, X_2, X_3, X_4) \sim \text{Multinomial}\left(50; \frac{1}{4}, \frac{1}{4}, \frac{1}{4}, \frac{1}{4}\right).$$

Conditional on $X_2 = 5$, the conditional distribution of X_1 is

- (a) Binomial(45, 1/4)
- (b) Binomial(45, 1/3)
- (c) Binomial(50, 1/3)
- (d) Poisson(15)

Forecast Problem 14. For the joint density

$$f(x, y) = \begin{cases} 2, & 0 < x < y < 1, \\ 0, & \text{otherwise,} \end{cases}$$

the probability $\mathbb{P}(X + Y < 1)$ is

- (a) $\frac{1}{4}$
- (b) $\frac{1}{3}$
- (c) $\frac{1}{2}$
- (d) $\frac{3}{4}$

P7: Distributions of Functions of Random Variables

Hit probability: 0.77. *Signal:* The recurring transformation quartet is normal-ratio $\rightarrow$ Cauchy, uniform-log $\rightarrow$ exponential, square maps, and gamma ratios $\rightarrow$ beta.

Forecast Problem 15. If X and Y are independent $N(0, 1)$ random variables, then the distribution of

$$\frac{X}{Y}$$

is

- (a) $N(0, 1)$
- (b) $\text{Cauchy}(0, 1)$
- (c) χ_1^2
- (d) $F(1, 1)$

Forecast Problem 16. If $X \sim U(0, 1)$ and $Y = -2 \log X$, then Y has

- (a) an exponential distribution with rate 2
- (b) an exponential distribution with rate $1/2$
- (c) a gamma distribution with shape $1/2$ and rate 1
- (d) a uniform distribution on $(0, 2)$

P8: Mathematical and Conditional Expectation

Hit probability: 0.84. *Signal:* Tower property, nonlinear binomial expectations, and variance of linear combinations.

Forecast Problem 17. Suppose

$$\mathbb{E}(X | Y) = Y^2 + 1, \quad \mathbb{E}(Y) = 1, \quad \text{Var}(Y) = 2.$$

Then $\mathbb{E}(X)$ is

- (a) 2
- (b) 3
- (c) 4
- (d) 5

Forecast Problem 18. If $X \sim \text{Binomial}(4, 1/2)$, then

$$\mathbb{E}\left(\frac{X}{X+1}\right)$$

is equal to

- (a) $\frac{31}{80}$
- (b) $\frac{49}{80}$
- (c) $\frac{5}{8}$
- (d) $\frac{3}{4}$

Forecast Problem 19. If

$$\text{Var}(X) = 4, \quad \text{Var}(Y) = 9, \quad \text{Cov}(X, Y) = 1,$$

then

$$\text{Var}(2X - 3Y)$$

is equal to

- (a) 73
- (b) 81
- (c) 85
- (d) 97

P9: Characteristic Functions, MGF, PGF and Compound Laws

Hit probability: 0.82. *Signal:* The compound-sum MGF is the strongest recent momentum signal and should be treated as a compulsory preparation item.

Forecast Problem 20. Let $N \sim \text{Poisson}(2)$, and let $X_1, X_2, \dots$ be independent $\text{Poisson}(3)$ random variables, independent of N . If

$$S_N = X_1 + \dots + X_N,$$

then the MGF of S_N is

- (a) $\exp\{2(e^{3t} - 1)\}$
- (b) $\exp\{3(e^{2t} - 1)\}$
- (c) $\exp\{2[\exp(3(e^t - 1)) - 1]\}$
- (d) $\exp\{3[\exp(2(e^t - 1)) - 1]\}$

Forecast Problem 21. If X and Y are independent $N(0, 1)$ random variables, the characteristic function of XY is

- (a) e^{-t^2}
- (b) $\frac{1}{1+t^2}$
- (c) $\frac{1}{\sqrt{1+t^2}}$
- (d) $e^{-2|t|}$

Forecast Problem 22. Let X be a non-negative integer-valued random variable with PGF

$$P(s) = \mathbb{E}(s^X).$$

For $|s| < 1$,

$$\sum_{n=0}^{\infty} s^n \mathbb{P}(X \leq n)$$

is equal to

- (a) $\frac{P(s)}{1-s}$
- (b) $\frac{P(s)}{1-s^2}$
- (c) $\frac{P(s^2)}{1-s}$
- (d) $\frac{P(\sqrt{s})}{1-s^2}$

P10: Modes of Convergence

Hit probability: 0.42. *Signal:* Constructed counterexamples separating almost-sure, probability, and mean-square convergence.

Forecast Problem 23. Let $U \sim U(0, 1)$, and define

$$X_n = \begin{cases} n, & 0 < U < \frac{1}{n^2}, \\ 0, & \text{otherwise.} \end{cases}$$

Then X_n converges to 0

- (a) almost surely and in mean square
- (b) almost surely but not in mean square
- (c) in mean square but not almost surely
- (d) neither almost surely nor in mean square

P11: Borel and Kolmogorov 0-1 Laws

Hit probability: 0.29. *Signal:* Low frequency, but the theorem statement is short and has high return per unit preparation time.

Forecast Problem 24. Let $A_1, A_2, \dots$ be independent events such that

$$\sum_{n=1}^{\infty} \mathbb{P}(A_n) = \infty.$$

Then

$$\mathbb{P}(A_n \text{ infinitely often})$$

is equal to

- (a) 0
- (b) $\frac{1}{4}$
- (c) $\frac{1}{2}$
- (d) 1

P12: Tchebycheff and Kolmogorov Inequalities

Hit probability: 0.46. *Signal:* Sample-size wording with “at least 0.95” remains the most likely form.

Forecast Problem 25. A population has standard deviation 1. By Tchebycheff’s inequality, how large must the sample size n be in order that

$$\mathbb{P}(|\bar{X} - \mu| < 0.5) \geq 0.95?$$

- (a) 40
- (b) 60
- (c) 79
- (d) 80

P13: Laws of Large Numbers and Central Limit Theorems

Hit probability: 0.57. *Signal:* Empirical CDF asymptotics, consistency, or a standardized sum with $\sqrt{n}$.

Forecast Problem 26. Let $F_n(x)$ be the empirical distribution function based on an iid sample from a continuous distribution with CDF F . For a fixed x satisfying $0 < F(x) < 1$,

$$\sqrt{n}(F_n(x) - F(x))$$

converges in distribution to

- (a) $N(0, 1)$
- (b) $N(F(x), F(x)(1 - F(x)))$
- (c) $N(0, F(x)(1 - F(x)))$
- (d) $N(x, x(1 - x))$

Answer Key for the Forecast Set

Q	1	2	3	4	5	6	7	8	9	10	11	12	13
Ans.	c	b	c	c	a	a	b	c	d	b	b	c	b
Q	14	15	16	17	18	19	20	21	22	23	24	25	26
Ans.	c	b	b	c	b	c	c	c	a	b	d	d	c

Final Revision Order

- Master one complete joint-density workflow: normalization, support sketch, marginal density, conditional density, conditional moment, and region probability.
- Memorize the Beta-I parameter equations:

$$\mathbb{E}(X) = \frac{\alpha}{\alpha + \beta}, \quad \text{Var}(X) = \frac{\alpha\beta}{(\alpha + \beta)^2(\alpha + \beta + 1)}.$$

- Memorize the compound-sum identity:

$$M_{S_N}(t) = M_N(\log M_X(t)).$$

- Drill the negative-binomial waiting-time form:

$$\mathbb{P}(T_n = m) = \binom{m-1}{n-1} p^n (1-p)^{m-n}.$$

- Rehearse the transformations

$$\frac{N(0, 1)}{N(0, 1)} \sim \text{Cauchy}(0, 1), \quad -\log U \sim \text{Exponential}(1), \quad \frac{G_1}{G_1 + G_2} \sim \text{Beta}(\alpha_1, \alpha_2).$$

- Keep the theorem-only blocks concise:

a.s. $\implies$ in probability $\implies$ in distribution,

$$\sum_n \mathbb{P}(A_n) < \infty \implies \mathbb{P}(A_n \text{ i.o.}) = 0,$$

and, under independence,

$$\sum_n \mathbb{P}(A_n) = \infty \implies \mathbb{P}(A_n \text{ i.o.}) = 1.$$